

# Sulthan Zahran Ma'ruf

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## Professional Experience

**Metatech** / Software product company building AI agents, observability tools, and messaging infrastructure

Remote, Indonesia

Apr 2026 – Present

### Software Engineer (Part-time)

- Built an AI chief of staff that handles a company's daily admin (email, meetings, cross-department updates), reachable from WhatsApp, web, or company chat, with per-role permissions.
- Built the messaging engine that moves messages between apps and chat platforms and the monitoring that catches failures in minutes, together generating **50B+ IDR in B2B revenue**.

**LG Sinarmas Technology Solutions** / IT Consultant company that handles Software-Hardware Integration on the biggest Smart Factory (EV Battery Factory) in South East Asia

Karawang, Indonesia

Apr 2024 – Present

### Software Engineer

- Acted as a primary liaison between Factory operations (Indonesian) and LG Engineering team (Korean), converting operator problem into clear, technical, actionable item for the main HQ team.
- Built an RAG pipeline (LangChain, PostgreSQL) for cross-language retrieval over Korean–English factory documentation.
- Wrote the software that runs the factory's robot movers: the carriers and conveyors that keep battery parts flowing between stations with no one pushing anything (.NET).
- Developed multiple internal applications (8+) to cut man hours, one of which saved **50 person-hours per day (React + Python based)**.
- Designed and delivered the team onboarding program. Trained **15+ Software Engineers** to be able to contribute in Smart Factory IT System and thus selected for a **2-month Korea deployment in LG Ensol Ochang Factory in South Korea**.

**PT Polychemie Asia Pacific Permai** / Chemical Factory that produces Presto Glue

Jakarta, Indonesia

Jan 2024 – Mar 2024

### Instrumentation Engineer (Projects)

- Gathered requirements for digitalization in a factory with fully manual business process (Python based)
- Prototyped GUI program that enables digitalization on these factories

**FAAN Laboratory** / The most productive Research Laboratory in Physics, Universitas Indonesia

Jakarta, Indonesia

Oct 2023 – Mar 2024

### Software Engineering Research Assistant

- Gathered analysis requirement from research staff and assistant professor, identify work bottleneck
- Translate these into a GUI applications that allows researcher to work interactively instead of needing separate session between data taking and analysis. Effectively improving the Synthesis-Testing cycle by almost ~800% (Matlab + Python based).

## Education

### Universitas Indonesia

B.Sc. Physics Instrumentation

Aug 2019 – Jul 2023

## Projects & Achievements

### Tim Robotika Universitas Indonesia's KRTMI Programming Lead

Lead one of the team in UI robotics team in a competition centered around Digital Twin Robots

### 4th Place team in IndySCC22

The only team from SouthEast Asia Competing in IndySCC, organized by IEEE Computer Society.

## Skills:

**AI / LLM:** RAG pipelines, Agentic AI Orchestration, MCP, LLM integration

**Languages:** Python, Go, C#, JavaScript, TypeScript, SQL, MATLAB

**Frameworks:** React, Flask, .NET, WPF

**Databases:** PostgreSQL, SQL Server, MySQL

**Systems:** REST APIs, real-time & distributed systems, ETL pipelines, MES/AMHS